INCH-POUND

MIL-DTL-17/128C
09 May 2019
SUPERSEDING
MIL-DTL-17/128B
w/AMENDMENT 2
25 February 2005

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/128-RG400

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

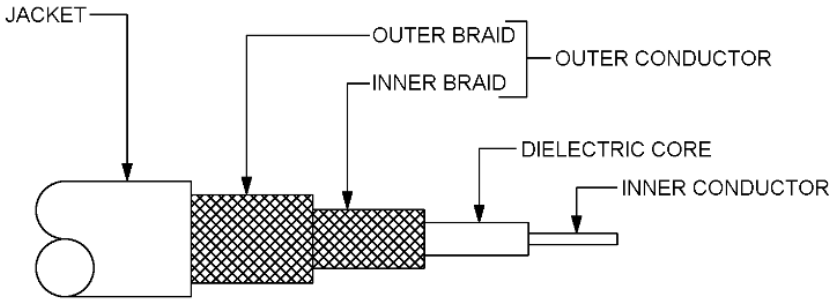


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Nineteen strands of silver-coated copper wire at .008 inch each. Overall diameter: .0384 inch ± .0010.
Dielectric core	Type F-1: Solid extruded PTFE. Diameter: .116 inch ± .005.
Outer conductor	Double braid of AWG size 36 silver-coated copper wire. Diameter: .171 inch maximum.
Inner braid	Coverage: 94.8% nominal Carriers: 16 Ends: 7 Picks/inch: 11.5 ± 10%
Outer braid	Coverage: 93.6% nominal Carriers: 16 Ends: 7 Picks/inch: 14.5 ± 10%
Jacket	Type IX: Diameter: .195 inch ± .005.

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ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 12.4 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -55°C to + 200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.91 ohm per 100 feet.

Elongation: 10 percent, minimum.

Tensile strength: Not applicable.

Engineering notes: This cable is useful in general purpose, high temperature applications (see connector series “TNC”, “BNC”, and “SMA” in accordance with MIL-PRF-39012).

REQUIREMENTS:

Dimensions, configuration and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable

Diameter: Applicable.

Out-of-roundness: Not Applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors: Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.

Electrical and mechanical:

Test frequency: 50 MHz to 12.4 GHz.

Continuity: Applicable.

Spark test: 2,000 V rms, +10%, -0%.

Voltage withstanding: 3,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ± 2 ohms.

RF transmission loss (attenuation): See figure 2.

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Standing wave ratio (return loss): See figure 3. Minimum cable length 7 feet.

Capacitance: 32 pF per foot, maximum.

Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Inductance: Not applicable.
Mechanically induced noise: Not applicable.
Time delay: Not applicable.
Aging stability: Not applicable.

Stress crack resistance: +230°C ± 5°C. Mandrel size: Seven times the jacket diameter.

Outer conductor integrity: Not applicable.

Cold bend: -55°C ± 2°C.

Dimensional stability: +200°C ± 5°C.

Inner conductor from core: 0.187 inch, maximum.

Inner conductor from jacket: 0.312 inch, maximum.

Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not Applicable.

Flame propagation: Not applicable.
Acid gas generation: Not applicable.
Halogen: Not applicable.
Fluid immersion: Not applicable.
Smoke index: Not applicable.
Toxicity index: Not applicable.
Durometer hardness: Not applicable.
Weathering: Not applicable.
Abrasion resistance: Not applicable.
Tear strength: Not applicable.
Heat distortion: Not applicable.
Physicals: Not applicable.

Tensile strength and elongation: Not applicable.

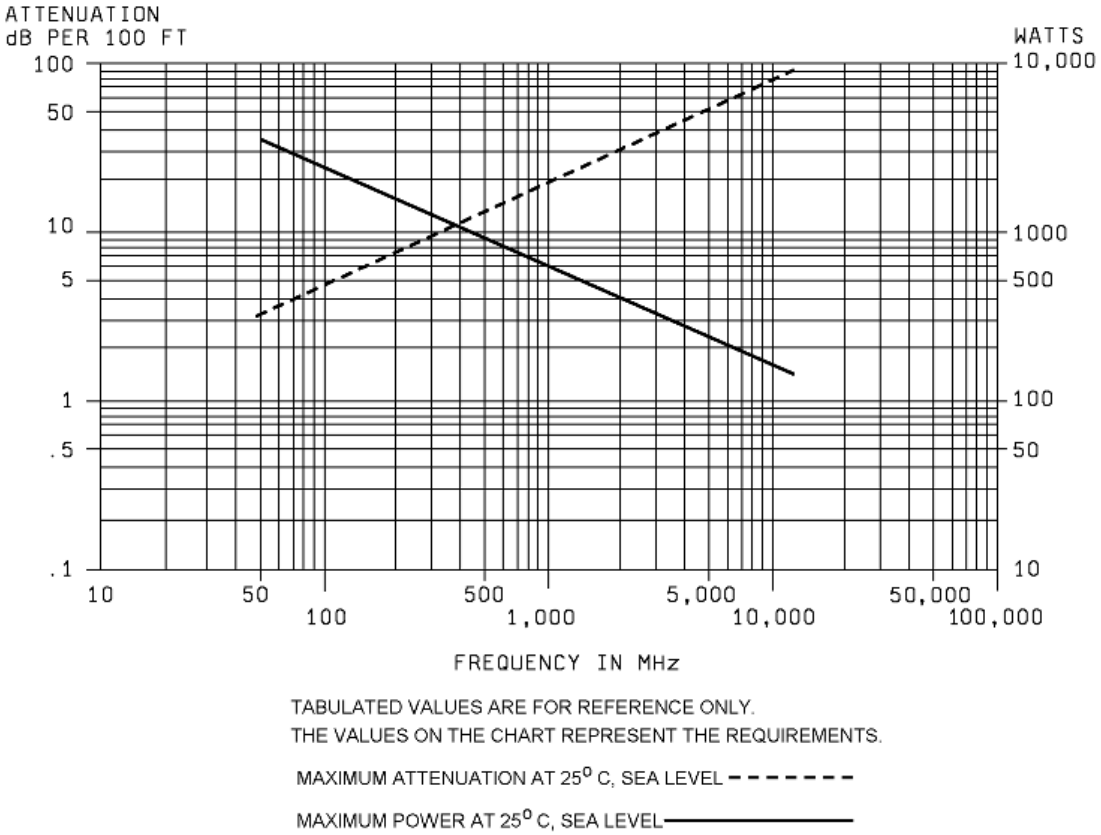
Weight: 5 pounds per 100 feet, maximum.

Marking: Applicable.

Workmanship: Applicable.

Part or Identifying Number (PIN): M17/128-RG400.

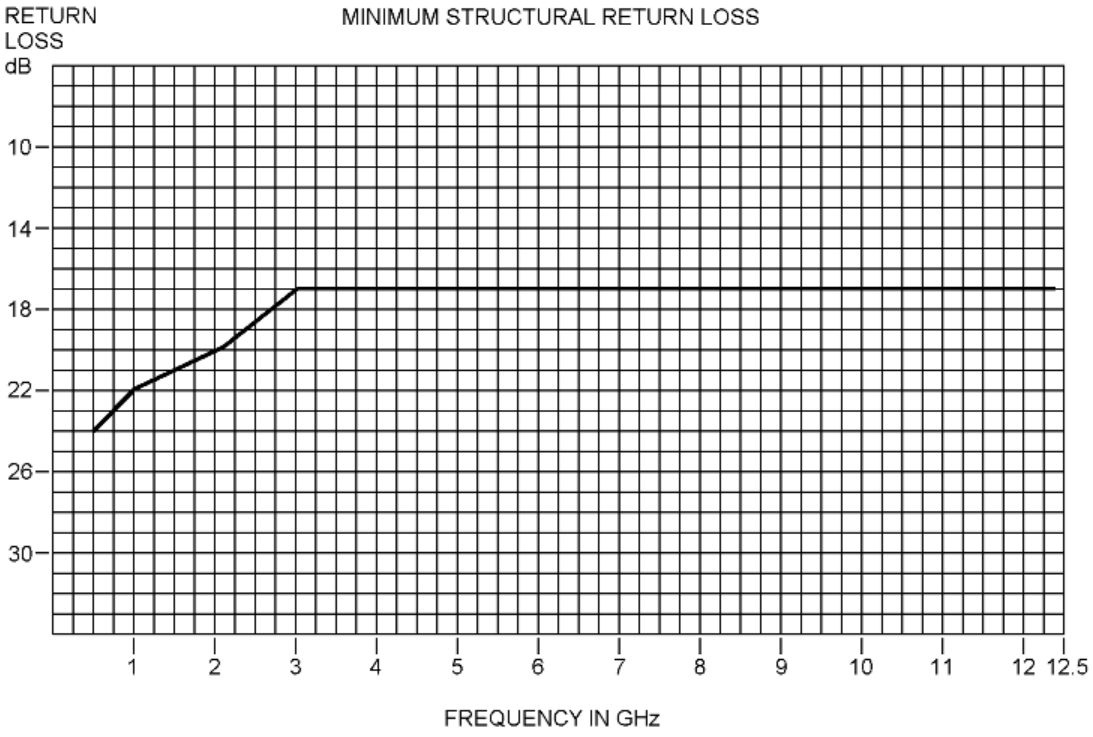
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Frequency MHz	Attenuation dB
50	3.2
100	4.5
400	10.5
1,000	17
3,000	38
10,000	78
12,400	90

FIGURE 2. Power rating and attenuation.

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TABULATED VALUES ARE FOR REFERENCE ONLY.
THE VALUES ON THE CHART REPRESENT THE REQUIREMENTS.

MHz	dB
50	25.5
400	23.8
1000	22
2000	20
3000	17
8000	17

SWR	RELECTION COEFFICIENT	RETURN LOSS Db
1.4985	.1995	14
1.4326	.1778	15
1.3767	.1585	16
1.3290	.1413	17
1.2880	.1259	18
1.2528	.1122	19
1.2222	.1000	20
1.1957	.0891	21
1.1726	.0794	22
1.1524	.0708	23
1.1347	.0631	24
1.1192	.0562	25
1.1055	.0501	26

FIGURE 3. Structural return loss.

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TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	N/A				
	Stress crack resistance	3.7.16	4.8.17	4	Per Table VIII	C
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C

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TABLE II. Qualification and Group A, B, and C Table - Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion tests	N/A				
	Smoke index	N/A				
	Toxicity Index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals (aged)	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19
DLA - CQ

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